

MATH 211 Analysis III — Problem Session 2

Problem 1

Show that $\mathbb{Q}$ is neither open nor closed in $\mathbb{R}$.

Solution. Recall that every open interval (a, b) with $a, b \in \mathbb{R}$, $a < b$ contains a rational and an irrational number. Let q be a rational number and let $\epsilon > 0$ be any real number. The open ball around q with radius ϵ , $\text{Br}(q, \epsilon)$, contains an irrational number, which implies that $\mathbb{Q}$ is not an open set in $\mathbb{R}$. To show that $\mathbb{Q}$ is not closed in $\mathbb{R}$ one must show that $\mathbb{R} \setminus \mathbb{Q}$ is not open in $\mathbb{R}$, which follows by a similar argument to the one we used for the fact that $\mathbb{Q}$ is not open in $\mathbb{R}$.

Problem 2

Find the interior and the closure of each of the following sets:

$$(a, b), \quad (a, b], \quad [a, b]$$

where $a, b \in \mathbb{R}$, $a < b$.

Solution. Let $x \in (a, b)$. The open ball centered at x with radius $\epsilon = \min\{x - a, b - x\}$ is contained in (a, b) , which implies that the interior of (a, b) is (a, b) . We don't have to check points outside of (a, b) because a point in the interior of a set must be in the set itself.

We now investigate the boundary of (a, b) . An interior point can't be in the boundary, thus it is sufficient to check the points in the complement of (a, b) . The points a and b are in the boundary because for every $\epsilon > 0$, $\text{Br}(a, \epsilon)$ and $\text{Br}(b, \epsilon)$ contain both points of (a, b) and points of the complement of (a, b) , namely $(-\infty, a] \cup [b, +\infty)$. If $x \in (-\infty, a] \cup [b, +\infty)$ and $x > b$ then the open ball around x with radius $x - b$ does not intersect (a, b) . Thus any point greater than b can't be in the boundary. The case $x < a$ is similar. Thus, the boundary of (a, b) is $\{a, b\}$. Finally, since the closure of a set is the union of its interior and boundary we get the closure of (a, b) as $[a, b]$.

The other two cases, $(a, b]$ and $[a, b]$, can be done similarly.

Problem 3

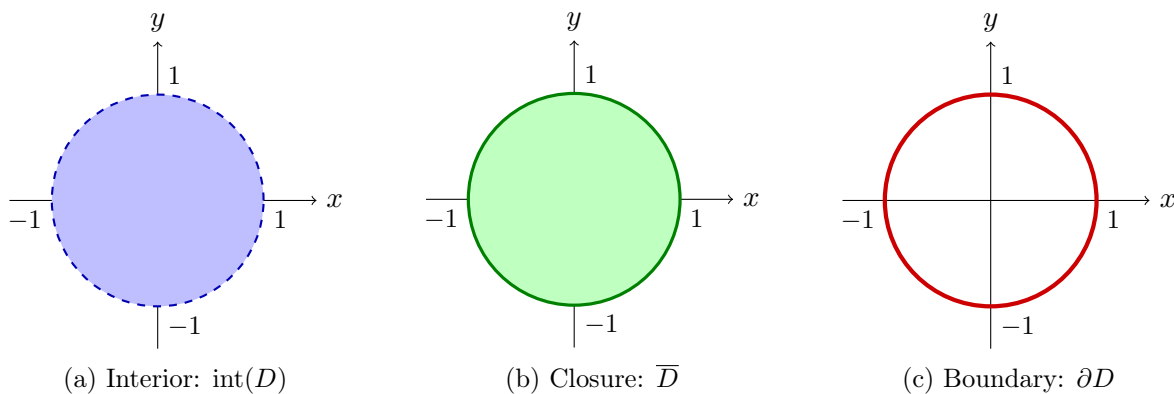
Let $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$. Find the interior, the closure and the boundary of D . Sketch each of them and shade the corresponding regions.

Solution. We find the interior of D . Let x be a point in D . We claim $\text{Br}(x, 1 - \|x\|)$ is a subset of D . Indeed, if $y \in \text{Br}(x, 1 - \|x\|)$ then:

$$\|y\| = \|y - x + x\| \leq \|y - x\| + \|x\| < 1 - \|x\| + \|x\| = 1.$$

The first inequality is by triangle inequality and the second inequality holds since $y \in \text{Br}(x, 1 - \|x\|)$. Thus, D is an open set and the interior of D is D itself. For the boundary of D , for all the points x satisfying $\|x\| = 1$ the ball around x contains both points of D and points of the complement of D . Thus such points are in the boundary. For the points x satisfying $\|x\| > 1$ the ball around x with radius $\|x\| - 1$ does not intersect D (verify it). Thus, the boundary of D is only the points with norm 1. Thus, the closure of D is the set

$$\overline{D} = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$



Problem 4

Show that the limit of a convergent sequence in $\mathbb{R}^n$ is unique.

Solution. Let x_k be a convergent sequence in $\mathbb{R}^n$. Suppose x and y are two limits of x_k . Let $\epsilon > 0$. By the definition of limit, there exist natural numbers N_1 and N_2 with the following properties:

- (i) $\|x_k - x\| < \epsilon$ for all $k > N_1$
- (ii) $\|x_k - y\| < \epsilon$ for all $k > N_2$.

Without loss of generality let $N_2 \geq N_1$. Then we have

$$\|x - y\| = \|x - y + x_k - x_k\| \leq \|x_k - y\| + \|x_k - x\| < 2\epsilon$$

for all $k > N_2$.

Since this holds for every $\epsilon > 0$, we get $\|x - y\| = 0$. By the properties of the norm function the only vector of zero norm is the zero vector. Thus $x - y = 0$, so $x = y$.

Problem 5

Prove that every Cauchy sequence in $\mathbb{R}^n$ converges.

Solution. In order to prove the statement we will employ the well-known theorem:

Theorem (Bolzano–Weierstrass). Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

The strategy of the proof is as follows: we first show that any Cauchy sequence x_k in $\mathbb{R}^n$ is bounded, then we conclude that this sequence has a convergent subsequence x_{k_j} (say converges to x) by using the Bolzano–Weierstrass theorem, and prove the theorem by showing x_k also converges to x .

Let x_k be a Cauchy sequence in $\mathbb{R}^n$. Let $\epsilon = 1$. By the definition of a Cauchy sequence there exists a natural number N such that $\|x_k - x_m\| < 1$ for every $k, m \geq N$. Thus $\|x_k\| \leq \|x_k - x_N\| + \|x_N\| < \|x_N\| + 1$ for every $k \geq N$. We managed to bound the terms of the sequence after the N 'th term but there might be finitely many points that are bigger than $\|x_N\| + 1$. We solve this issue by the following argument:

Let

$$M = \max\{\|x_1\|, \|x_2\|, \dots, \|x_{N-1}\|, \|x_N\| + 1\}.$$

It follows that $\|x_k\| \leq M$ for every k , thus x_k is a bounded sequence. Observe that we also proved that in order to show the boundedness of a sequence, we can ignore finitely many terms since we

can take the maximum over finitely many terms. Now by the Bolzano–Weierstrass theorem x_k has a subsequence x_{k_j} that converges (say to x). We prove x_k also converges to x .

Let $\epsilon > 0$. Then

$$\|x_k - x\| = \|x_k - x_{k_j} + x_{k_j} - x\| \leq \|x_k - x_{k_j}\| + \|x_{k_j} - x\| < \epsilon$$

for large enough k, j . $\|x_k - x_{k_j}\| < \frac{\epsilon}{2}$ as k, j go to $+\infty$ because x_k is a Cauchy sequence. $\|x_{k_j} - x\| < \frac{\epsilon}{2}$ because x_{k_j} converges to x .

Exercise 1. Show that every convergent sequence in $\mathbb{R}^n$ is a Cauchy sequence.

Exercise 2. Find a Cauchy sequence of rational numbers that does not converge to a rational number.